Robust Control Architectures for Trajectory Tracking of Skid-Steer Autonomous UGVs for Power Substation Inspection

A thesis submitted in partial fulfillment of the requirements for the award of the degree of

B.Tech.

in

Instrumentation and Control Engineering

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CERTIFICATE OF COLLABORATION

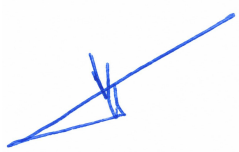
This is to certify that the project titled **"Robust Control Architectures for Trajectory Tracking of Skid-Steer Autonomous UGVs for Power Substation Inspection"** represents the work carried out by

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in collaboration with the **Smart Materials, Structures and Systems (SMSS) Laboratory, Indian Institute of Technology Kanpur**, with technical support from the **Central Power Research Institute (CPRI), India**, during the academic year 2025–2026, under the supervision of **Prof. Bishakh Bhattacharya**.

Chhavi Singh is a bona fide student of the **National Institute of Technology Jalandhar**, pursuing a Bachelor of Technology degree in Instrumentation and Control Engineering. Her contributions to this project demonstrate proficiency in control theory and the development and simulation of MATLAB/Simulink-based digital twins for autonomous ground vehicle systems.

The certificate is issued in recognition of her collaborative research contributions to the aforementioned project.



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Contents

0.1 Introduction . . . 3
0.2 Literature Review . . . 4
0.3 Methodology . . . 5
0.3.1 Kinematic Model . . . 6
0.3.2 Dynamic Model . . . 6
0.3.3 Tracking Error Formulation . . . 8
0.4 Simulation Setup . . . 9
0.5 Control Architectures . . . 11
0.5.1 Method I: Lyapunov-Based Kinematic Controller with LESO-STA Dynamic Compensation . . . 11
0.5.2 Method II: Gain-Scheduled Backstepping with Dynamic ICR Compensation . . . 13
0.5.3 Method III. Nonlinear Model Predictive Control (NMPC) with ADRC 14
0.6 Results and Discussion . . . 16
0.6.1 Lyapunov + STA-SMC . . . 16
0.6.2 Backstepping + STA-SMC . . . 16
0.6.3 NMPC + ADRC + LESO . . . 16
0.6.4 Robustness Under Discontinuous Trajectories . . . 16
0.6.5 Tracking Performance . . . 17
0.6.6 Disturbance and State Estimation . . . 19
0.6.7 Discussion . . . 19

A STABILITY ANALYSIS **20**
A.0.1 Mathematical Preliminaries . . . 20
A.0.2 Stability of the LESO Estimation Error . . . 21
A.0.3 Input-to-State Stability of the NMPC Closed-Loop System . . . 22
A.0.4 Composite Stability of the Cascaded NMPC-LESO-ADRC Framework . . . 23

0.1 Introduction

Mobile robots are widely used in industrial inspection, surveillance, logistics, and autonomous navigation. Their motion capabilities depend strongly on the chosen **locomotion architecture**, which determines how the robot generates forces and moves on the ground. The most common wheeled robot types include differential drive, Ackermann steering, skid–steer platforms, and omnidirectional robots.

Wheeled Mobile Robot Architectures

Table 1: Comparison of Mobile Robot Locomotion Architectures

Robot Type	**Steering Method**	**Sideways Motion**	**Typical Use**
Differential Drive	Left–right wheel speed	No	Indoor robots, UGVs
Ackermann Steering	Mechanical linkage	No	Cars, high-speed driving
Skid–Steer	Left/right side speed	No (skid)	Off-road, construction
Omnidirectional	Swedish/Mecanum wheels	Yes	Precise indoor motion

Non-Holonomic vs Holonomic Motion

Most ground robots (differential, Ackermann, skid–steer) are **non-holonomic**: they cannot move sideways and must rotate to change direction. Omnidirectional robots, using Swedish or spherical wheels, are **holonomic**, capable of instantaneous motion in any direction but requiring more complex actuation. **Holonomic robots** offer high manoeuvrability but have complex design. **Non-holonomic robots** are simple and robust, but less manoeuvrable.

Skid–steer robots achieve turning by driving the two sides at different velocities. However, steering involves **lateral skidding**, making ideal no-slip kinematic assumptions invalid. This introduces highly nonlinear friction forces and makes motion prediction difficult using kinematics alone.

0.2 Literature Review

Industrial inspection increasingly relies on skid-steer unmanned ground vehicles (UGVs) due to their structural simplicity and maneuverability on uneven terrain. However, precise trajectory tracking remains challenging as wheel slip, terrain induced disturbances, and complex wheel-ground interactions can degrade navigation performance, particularly during sharp maneuvers.

Extensive research has addressed skid-steer mobile robot (SSMR) tracking under these non-ideal conditions. Early studies focused on kinematic trajectory tracking of skid-steer robots. [1] developed a Lyapunov-based nonlinear trajectory tracking controller using a unicycle approximation for low-speed operation. [2] proposed a virtual orientation-based controller that decouples the nonlinear kinematics into first-order systems, while [3] introduced a friction-based kinematic model that predicts wheel slip using wheel-ground interaction dynamics to improve trajectory prediction. Subsequently, nonlinear robust control strategies were developed. [4] proposed a backstepping-based path-following controller with simplified steering dynamics. [5] combined Sliding Mode Control with an Extended State Observer (ESO) to estimate disturbances and enhance tracking robustness. Recently, Model Predictive Control (MPC) and nonlinear MPC (NMPC) variants (e.g., slip-aware, tube-based and robust constrained) have gained traction for their constraint-handling and optimization capabilities [6, 7, 8], with comparative studies confirming MPC's superiority over purely kinematic methods under slip conditions [9]. Modern frameworks often integrate multiple control layers. For instance, [10] combines kinematic tracking with an observer-based SMC dynamic controller for differential drive robots.

Despite these advancements, many tracking methods still use simplified kinematics that omit slip and actuator dynamics. The integration of disturbance estimation with nonlinear and predictive control under wheel–ground uncertainties remains limited and comprehensive comparative studies of cascaded control architectures for SSMRs remain scarce. To bridge these gaps, this paper provides two primary contributions:

1. A comparative evaluation of three cascaded tracking architectures for SSMRs: Lyapunov-LESO-ADRC-SMC, adaptive backstepping (with ICR adaptation)-LESO-ADRC-SMC, and NMPC-LESO-ADRC.

2. Validation on a high-fidelity Simscape Multibody framework to evaluate wheel-ground interactions and slip dynamics across smooth and discontinuous trajectories.

0.3 Methodology

System Architecture

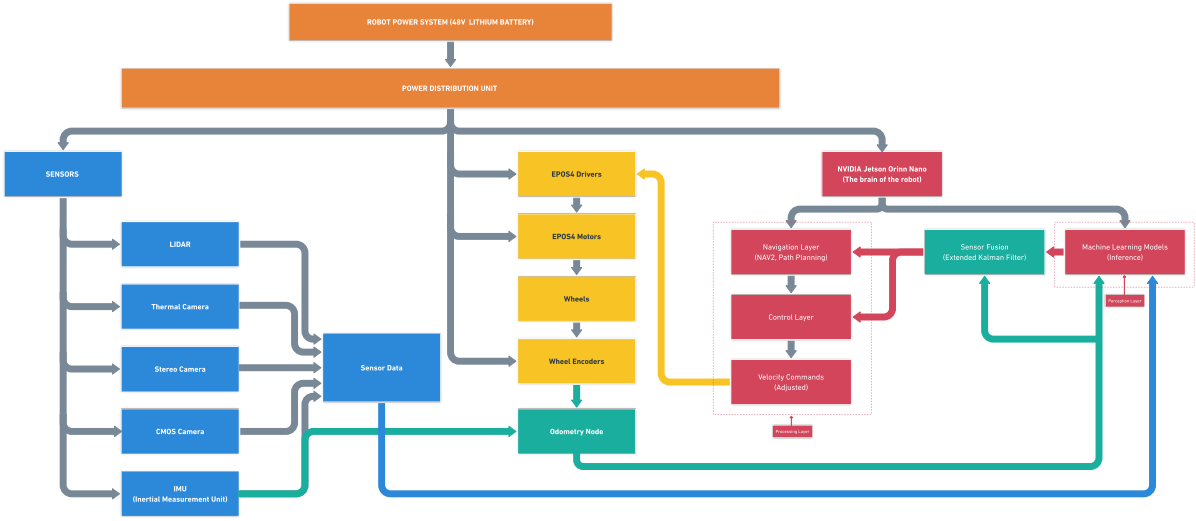


Figure 1: System architecture of the robotic platform illustrating the power distribution, sensing subsystem, perception and processing layers, and control pipeline enabling autonomous navigation.

The robotic platform architecture, shown in Fig. 1, integrates sensing, perception, processing, and control modules for autonomous navigation. A 48 V lithium battery powers the system through a centralized power distribution unit. The sensing layer comprises LiDAR, thermal, stereo, and CMOS cameras, along with an inertial measurement unit (IMU). Sensor measurements are fused using an Extended Kalman Filter (EKF) to estimate the robot state. Processing and inference are performed on an NVIDIA Jetson Orin Nano, which executes machine learning models for environmental perception. The navigation stack (NAV2) generates velocity commands based on path planning and trajectory optimization. These commands are executed by EPOS4 motor drivers controlling the wheel motors, while wheel encoders provide odometry feedback for closed-loop motion control.

The derivation of the kinematic and dynamic relations is based on the following assumptions: the centre of mass lies on the robot's body reference line and remains fixed; the vehicle is perfectly rigid and moves on a horizontal plane, with all four wheels maintaining continuous contact with the ground through point contacts. The two wheels on each side are assumed to rotate at the same angular velocity. The robot operates at low speeds, below $1,\mathrm{m/s}$, and longitudinal wheel slip is neglected. Furthermore, lateral tyre forces are assumed to depend on the vertical load acting on the wheels.

0.3.1 Kinematic Model

Following the nonholonomic formulation in [10], the skid-steer robot is approximated as an equivalent differential-drive model operating on a planar surface under pure-rolling and no-slip conditions. To prevent mathematical singularities in the kinematic transformation and enable multi-directional tracking control, the reference point is defined at a longitudinal lookahead offset d ahead of the geometric wheel axis center, where $p(t) = [\, x(t) \quad y(t) \quad \theta(t) \,]^T$ and $p_r(t) = [\, x_r(t) \quad y_r(t) \quad \theta_r(t) \,]^T$ denote the current and reference pose vectors of the robot, respectively. The skid-steer mobile robot satisfies the nonholonomic constraint

$$-\dot{x}(t)\sin\theta(t) + \dot{y}(t)\cos\theta(t) = 0, \tag{1}$$

which indicates that the robot cannot move laterally in its body-fixed frame. Accordingly, the kinematic model of the robot is expressed as

$$\begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\theta(t) & -d\sin\theta(t) \\ \sin\theta(t) & d\cos\theta(t) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_x(t) \\ \omega(t) \end{bmatrix} = S(p(t))\, v_o(t), \tag{2}$$

where $v_o(t) = [\, v_x(t) \quad \omega(t) \,]^T$ contains the linear velocity $v_x(t)$ and angular velocity $\omega(t)$ of the robot.

0.3.2 Dynamic Model

The constrained dynamic model of SSMRs is given by

$$M(p)\ddot{p} + C(p,\dot{p})\dot{p} + d_p = B(p)\tau - A^T(p)\lambda, \tag{3}$$

where $M(p) \in \mathbb{R}^{3\times3}$ is the positive-definite inertia matrix, $C(p,\dot{p}) \in \mathbb{R}^{3\times3}$ represents Coriolis and centripetal effects, $B(p) \in \mathbb{R}^{3\times2}$ is the input transformation matrix, $\tau = [\tau_l, \tau_r]^T$ denotes the wheel torque vector, $d_p = [d_{px}, d_{py}, d_{p\theta}]^T$ represents unknown external disturbances, and $A^T(p)\lambda$ corresponds to the nonholonomic constraint forces. Differentiating (2) yields

$$\ddot{p} = \dot{S}(p)v_o + S(p)\dot{v}_o. \tag{4}$$

Substituting into (3) and pre-multiplying by $S^T(p)$ eliminates the constraint force term due to the property $S^T(p)A^T(p) = 0$, resulting in

$$\bar{M}\dot{v}_o + \bar{C}v_o + \bar{d} = \bar{B}\tau, \tag{5}$$

where $\bar{M} = S^T M S$, $\bar{C} = S^T M \dot{S} + S^T C S$, $\bar{B} = S^T B$, and $\bar{d} = -S^T d_p$ are the reduced in-

ertia, Coriolis, input, and disturbance matrices obtained after projection onto the admissible velocity space. The disturbance vector $\bar{d} = [d_1, d_2]^T$ represents the projection of the unknown spatial disturbances onto the admissible velocity manifold. Here, $v = [\, v_l \;\; v_r \,]^T$ denotes the wheel velocity vector consisting of the left and right wheel velocities The Coriolis coupling, rolling resistance, wheel–ground interaction, parameter uncertainties, and external disturbances are subsequently lumped into a generalized disturbance vector.

$$\Delta(t) = \bar{C}(p, \dot{p})\, v_o + \bar{d}(t), \tag{6}$$

where $\bar{d}$ represents the projected external disturbances after eliminating the nonholonomic constraint. Hence, the reduced dynamics are expressed as

$$\bar{M}\dot{v}_0 + \Delta(t) = \bar{B}\tau. \tag{7}$$

Using the reduced inertia and input matrices

$$\bar{M} = \begin{bmatrix} m & 0 \\ 0 & I_z - md^2 \end{bmatrix}, \qquad \bar{B} = \frac{1}{r}\begin{bmatrix} 1 & 1 \\ b & -b \end{bmatrix}, \tag{8}$$

Defining the normalized disturbances as $d_1 = -\dfrac{\Delta_1}{m}$ and $d_2 = -\dfrac{\Delta_2}{I_r - md^2}$, the reduced dynamics can be rewritten as

$$\dot{v}_x = \frac{1}{mr}\,(\tau_l + \tau_r) + d_1, \tag{9}$$

$$\dot{\omega} = \frac{b}{(I_z - md^2)r}\,(\tau_l - \tau_r) + d_2. \tag{10}$$

The relationship between the chassis velocity and the wheel velocities is

$$\begin{bmatrix} v_l \\ v_r \end{bmatrix} = \begin{bmatrix} 1 & b \\ 1 & -b \end{bmatrix}\begin{bmatrix} v_x \\ \omega \end{bmatrix}, \tag{11}$$

Substituting (9) and (10) into Differentiated form of (11) gives

$$\dot{v}_l = \frac{1}{mr}(\tau_l + \tau_r) + \frac{b^2}{(I_z - md^2)r}(\tau_l - \tau_r) + D_l, \tag{12}$$

$$\dot{v}_r = \frac{1}{mr}(\tau_l + \tau_r) - \frac{b^2}{(I_z - md^2)r}(\tau_l - \tau_r) + D_r, \tag{13}$$

where

$$D_l = d_1 + bd_2, \qquad D_r = d_1 - bd_2. \tag{14}$$

Rearranging (12) and (13) according to the input torques yields

$$\dot{v}_l = \alpha_l \tau_l + \beta_l \tau_r + D_l, \tag{15}$$

$$\dot{v}_r = \alpha_r \tau_r + \beta_r \tau_l + D_r, \tag{16}$$

where $\alpha_l = \alpha_r = \dfrac{1}{mr} + \dfrac{b^2}{(I_r - md^2)r}$ and $\beta_l = \beta_r = \dfrac{1}{mr} - \dfrac{b^2}{(I_r - md^2)r}$. Since the exact control effectiveness is affected by unmodeled dynamics, the true input gain is generally unknown. Following the ADRC philosophy[11],where the nominal input gain is selected as based on the nominal control effectiveness and fine-tuned experimentally.

Accordingly, each wheel is represented by the standard first-order ADRC model

$$\dot{v}_i = b_0 \tau_i + f_i(t), \qquad i \in \{l, r\}, \tag{17}$$

where

$$f_i(t) = D_i + (\alpha_i - b_0)\tau_i + \beta_i \tau_j, \qquad i \neq j, \tag{18}$$

is the generalized disturbance that lumps wheel–ground interaction, slip, inter-wheel coupling, rolling resistance, parameter uncertainties, unmodeled dynamics, external disturbances, and input-gain mismatch. The LESO estimates $f_i(t)$ online for ADRC-based compensation.

0.3.3 Tracking Error Formulation

The kinematic tracking error model transforms the state error vector into the robot's local coordinate frame following the canonical framework [9]. To handle non-holonomic constraints and ensure control stability, the tracking point is shifted from the wheel axle center to an off-axle lookahead point at a longitudinal displacement d.

Let $(x_r,\ y_r,\ \theta_r)$ denote the reference pose, $(v_{xr},\ \omega_r)$ the corresponding reference linear and angular velocities, and $(v_{xa},\ \omega_a)$ the actual linear and angular velocities of the robot. Then, the kinematic tracking errors can be computed as

$$\begin{bmatrix} x_e \\ y_e \\ \theta_e \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r - x \\ y_r - y \\ \theta_r - \theta \end{bmatrix}. \tag{19}$$

The time derivatives of the tracking error states (x_e, y_e, θ_e) are formulated as:

$$\begin{aligned} \dot{x}_e &= y_e \omega_a + v_{xr} \cos\theta_e - d\omega_r \sin\theta_e - v_{xa} \\ \dot{y}_e &= -x_e \omega_a + v_{xr} \sin\theta_e + d\omega_r \cos\theta_e - d\omega_a \\ \dot{\theta}_e &= \omega_r - \omega_a \end{aligned} \tag{20}$$

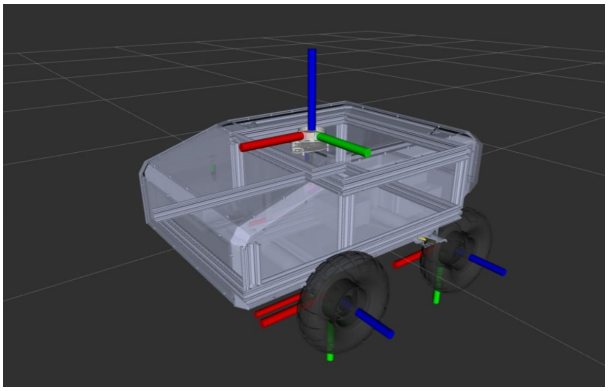


Figure 2: URDF representation of the robot showing the body links and wheel joints used for simulation.

0.4 Simulation Setup

The robot model developed in URDF was imported into MATLAB using the `smimport` utility, which converts the URDF description into an equivalent Simscape Multibody model Fig 3. This process automatically generates the rigid bodies, joint structures, and coordinate transformations corresponding to the kinematic structure of the robot.

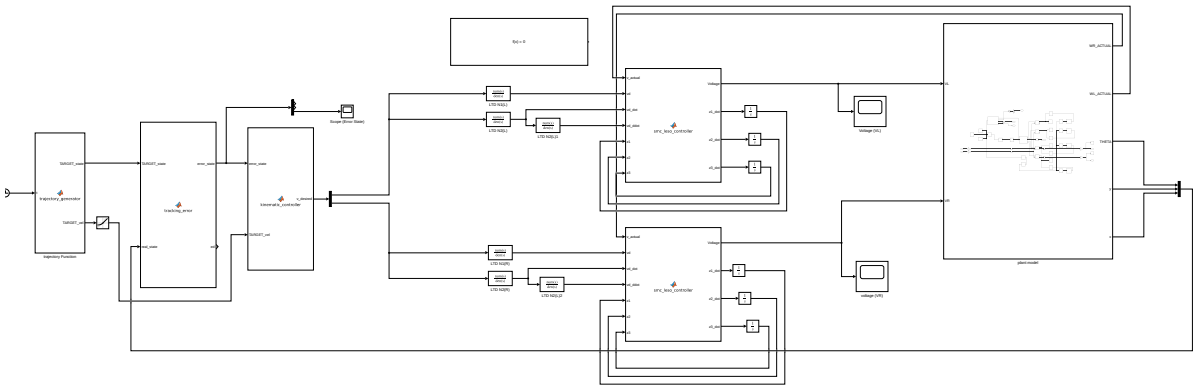


Figure 3: Simulink model of the system.

After importing the URDF, the generated Simscape model was modified to enable interaction with the environment. A ground plane was modeled using a solid body representing the terrain surface. To simulate the physical interaction between the robot and the ground, the *Spatial Contact Force* block was used. This block models the normal contact force and friction forces between contacting bodies, allowing realistic ground reaction forces to be generated during simulation.

The parameters used in the spatial contact force model are summarized in Table 2. These parameters define the stiffness, damping, and friction properties governing the contact interaction between the robot and the ground.

Parameter	Symbol	Value
Normal Stiffness	k_n	$1 \times 1e^5$ N/m
Normal Damping Coefficient	c_n	100 N/(m/s)
Static Friction Coefficient	μ_s	0.9
Dynamic Friction Coefficient	μ_d	0.8
Transition Velocity	v_t	0.01 m/s

Table 2: Spatial Contact Force Parameters Used for Wheel–Ground Interaction

To allow the robot to move freely in the environment, a *6-DOF Joint* was introduced between the robot base and the world frame. This joint removes rigid constraints and permits translation and rotation of the robot body in all spatial directions. Consequently, the robot can respond dynamically to wheel actuation and contact forces generated by the terrain interaction.

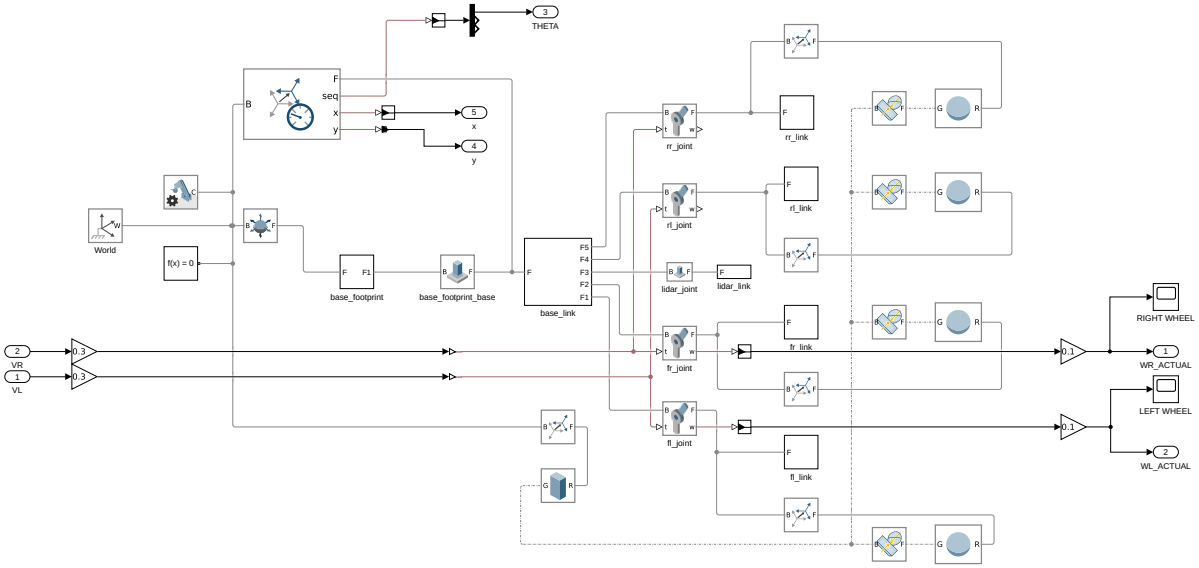


Figure 4: Simulink model of the plant

0.5 Control Architectures

0.5.1 Method I: Lyapunov-Based Kinematic Controller with LESO-STA Dynamic Compensation

Following the Lyapunov-based trajectory tracking framework proposed in [9], the desired linear and angular velocities are selected as

$$v_c = k_1 \left(x_e + d(1 - \cos\theta_e)\right) + v_r \cos\theta_e \tag{21}$$

$$\omega_c = \omega_r + k_2(y_e - d\sin\theta_e)\cos\left(\frac{\theta_e}{2}\right) + k_3 \sin\left(\frac{\theta_e}{2}\right) \tag{22}$$

Wheel velocity mapping:

$$v_l = v_c - b\omega_c, \quad v_r = v_c + b\omega_c \tag{23}$$

The variables v_c and ω_c denote the commanded linear and angular velocities, respectively, while v_r and ω_r represent the corresponding reference velocities. The control gains are denoted by k_1, k_2, and k_3. The look-ahead distance is represented by d, and b denotes the half-track width of the vehicle.

Assumption 1: The kinematic controller is designed using the nominal nonholonomic model. Slip, ICR variations, wheel scrubbing, and wheel-ground interactions are treated as disturbances and compensated in the dynamic layer.

Dynamic Layer

Linear Tracking Differentiator (LTD) A Linear Tracking Differentiator (LTD) is employed to generate smooth wheel velocity references [9]:

$$\dot{n}_1 = n_2, \tag{24}$$

$$\dot{n}_2 = -\zeta^2(n_1 - v_d) - 2\zeta n_2, \tag{25}$$

where n_1 and n_2 denote the filtered wheel velocity reference and its derivative, respectively.

Linear Extended State Observer (LESO) The reduced dynamics are

$$\dot{v} = f(t) + b_0 u, \tag{26}$$

where v denotes the measured wheel velocity and f represents the total lumped disturbance.

Let,

$$x_1 = v, \qquad x_2 = f, \tag{27}$$

the second-order LESO is The LESO is formulated as

$$\dot{\hat{x}}_1 = \hat{x}_2 + b_0 u + \gamma_1 l \left(y - \hat{x}_1\right), \tag{28}$$

$$\dot{\hat{x}}_2 = \gamma_2 l^2 \left(y - \hat{x}_1\right), \tag{29}$$

where $y = v$ is the measured wheel velocity , b_0 is the nominal input gain, and l is the observer bandwidth. The observer gains are chosen as $\gamma_1 = 2$ and $\gamma_2 = 1$.

Assumption 2: The generalized disturbance satisfies

$$|\dot{f}(t)| \leq \bar{f}, \tag{30}$$

where $\bar{f} > 0$ is a constant.

ADRC-Based Sliding Mode Controller Following the ADRC framework [11], the LESO-estimated generalized disturbance is actively compensated, while a smooth sliding mode term improves robustness against residual estimation errors and unmodeled dynamics. Unlike conventional ADRC, which employs Linear State Error Feedback (LSEF), the proposed approach replaces the LSEF with a smooth sliding mode controller.

The wheel-speed tracking error is defined as

$$e_i = v_i - n_{1,i}, \qquad i \in \{l, r\}, \tag{31}$$

where v_i is the measured wheel velocity and $n_{1,i}$ is the filtered wheel velocity reference generated by the Linear Tracking Differentiator (LTD).

$$s_i = e_i. \tag{32}$$

$$u_0 = n_{2,i} + k_p e_i, \tag{33}$$

where $n_{2,i} = \dot{n}_{1,i}$ is the derivative of the filtered reference generated by the LTD and $k_p > 0$ is the proportional tracking gain. Using the LESO disturbance estimate $\hat{f} = z_2$, the equivalent control becomes

$$u_{eq} = u_0 - \hat{f} = u_0 - z_2, \tag{34}$$

To improve robustness against disturbance estimation errors and wheel–ground interaction effects, the smooth sliding mode correction is introduced as

$$u_{sw} = -k_1 \tanh\left(\frac{s_i}{\phi}\right) - k_3 s_i, \tag{35}$$

where $k_1, k_3 > 0$ are the sliding mode gains and $\phi > 0$ is the boundary-layer thickness. The hyperbolic tangent function provides a continuous approximation of the signum function, thereby reducing chattering.

Finally, the wheel torque command is computed as

$$\tau_i = \frac{u_{eq} + u_{sw}}{b_0}, \tag{36}$$

where b_0 is the nominal input gain.

0.5.2 Method II: Gain-Scheduled Backstepping with Dynamic ICR Compensation

Kinematic Layer

The gain-scheduled backstepping controller with dynamic ICR compensation is adopted from [12]. The posture tracking errors are defined in the robot body-fixed frame as (31) Backstepping-based control law:

$$v_c = v_r \cos(\theta_e) + k_1 x_e \tag{37}$$

$$\omega_c = \omega_r + k_2(|v_r| + \epsilon) y_e + k_3(|v_r| + \epsilon)\sin(\theta_e) \tag{38}$$

Adaptive ICR compensation:

$$d(t) = X_{icr}^{max} \exp(-\alpha|\kappa| v_r^2) \tag{39}$$

$$\kappa = \frac{\omega_r}{v_r} \tag{40}$$

where k_1, k_2, and k_3 are positive control gains and $\epsilon > 0$ is a small constant introduced to preserve control effectiveness at low speeds.

The curvature-dependent scheduling law is defined by the curvature κ and the time-varying lookahead distance $d(t)$, where $X_{ICR,\max}$ denotes the maximum ICR offset and α is a tuning parameter.As the curvature or vehicle speed increases, the effective ICR shifts toward the robot, compensating for wheel slip and nonlinear skid-steering effects. The desired wheel velocities are obtained as

$$v_l = v_c - d(t)\omega_c - b\omega_c \tag{41}$$

$$v_r = v_c - d(t)\omega_c + b\omega_c \tag{42}$$

where b is half of the track width. The corresponding wheel angular velocity references

are given by

$$\omega_{ld} = \text{sat}\left(\frac{v_l}{r}\right), \quad \omega_{rd} = \text{sat}\left(\frac{v_r}{r}\right) \tag{43}$$

where r denotes the wheel radius and $\text{sat}(\cdot)$ represents the actuator saturation function.

Dynamic Layer

The dynamic control strategy is identical to that used in Method I. The wheel velocity references generated by the gain-scheduled backstepping controller are tracked using the LESO assisted hybrid ADRC-SMC framework presented in Section 0.5.1.

0.5.3 Method III. Nonlinear Model Predictive Control (NMPC) with ADRC

Kinematic Control Design (NMPC-Based)

To account for system constraints while optimizing trajectory tracking performance, a Nonlinear Model Predictive Control (NMPC) strategy is employed at the kinematic layer using posture tracking errors defined in the robot body-fixed frame. The corresponding nonlinear error dynamics are given by (31).
The NMPC computes the optimal control sequence by minimizing a finite-horizon cost function:

$$J = \sum_{k=1}^{N} \left(x_{e,k}^T Q x_{e,k} + u_k^T R u_k\right) \tag{44}$$

where:

$$x_{e,k} = \begin{bmatrix} x_e & y_e & \theta_e \end{bmatrix}^T \tag{45}$$

and

$$u_k = \begin{bmatrix} v & \omega \end{bmatrix}^T \tag{46}$$

The weighting matrices are chosen as $Q = \text{diag}(20, 20, 1)$ and $R = \text{diag}(5, 2)$, with a finite prediction horizon of $N = 10$ and a sampling interval of $\Delta t = 0.1\,\text{s}$.

Control Constraints

The admissible control inputs are constrained as:

$$-0.3 \leq v \leq 0.6 \tag{47}$$

$$-0.7 \leq \omega \leq 0.7 \tag{48}$$

These constraints ensure bounded wheel velocities, safe operation within actuator limits, smooth trajectory execution, and improved numerical stability during simulation and control.

Dynamic layer

The wheel dynamics are modeled as a second-order uncertain nonlinear system:

$$\dot{\omega} = f(t) + b_0 u \tag{49}$$

where ω is the wheel angular velocity, $f(t)$ is the lumped disturbance term, u is the wheel torque input, and b_0 is the nominal plant gain. The lumped disturbance accounts for terrain interaction, wheel slip, unmodeled friction, and external disturbances.

Linear Extended State Observer (LESO)

The LESO estimates the wheel angular velocity and lumped disturbance in real time:

$$\dot{z}_1 = z_2 - \beta_1(z_1 - \omega) + b_0 u \tag{50}$$

$$\dot{z}_2 = -\beta_2(z_1 - \omega) \tag{51}$$

where z_1 and z_2 denote the estimated wheel velocity and lumped disturbance, respectively, β_1 and β_2 are the observer gains, and b_0 is the nominal control gain. The observer gains are set to $\beta_1 = 60$ and $\beta_2 = 250$.

Active Disturbance Rejection Control (ADRC)

The wheel velocity tracking error is defined as

$$e = \omega_{\text{ref}} - \omega \tag{52}$$

and the ADRC control law is given by

$$u = \frac{1}{b_0}\left(K_p e + \dot{\omega}_{\text{ref}} - z_2\right) \tag{53}$$

where ω_{ref} and $\dot{\omega}_{\text{ref}}$ denote the desired wheel velocity and reference acceleration, respectively, K_p is the proportional tracking gain, and z_2 is the disturbance estimate provided by the LESO. The reference acceleration feedforward term improves transient response, reduces tracking lag, ensures smoother tracking, and enhances disturbance rejection.

0.6 Results and Discussion

0.6.1 Lyapunov + STA-SMC

The Lyapunov-based controller achieved asymptotic convergence of posture tracking errors and bounded wheel velocity tracking under nonlinear disturbances. The LESO estimated disturbances arising from wheel slip and terrain interaction, while the Super-Twisting Sliding Mode Controller (STA-SMC) ensured robustness against model uncertainties. In Fig 6c sharp curvature changes induced transient oscillations and minor chattering-like behavior in wheel velocity responses, reflecting the controller's limited predictive capability.

0.6.2 Backstepping + STA-SMC

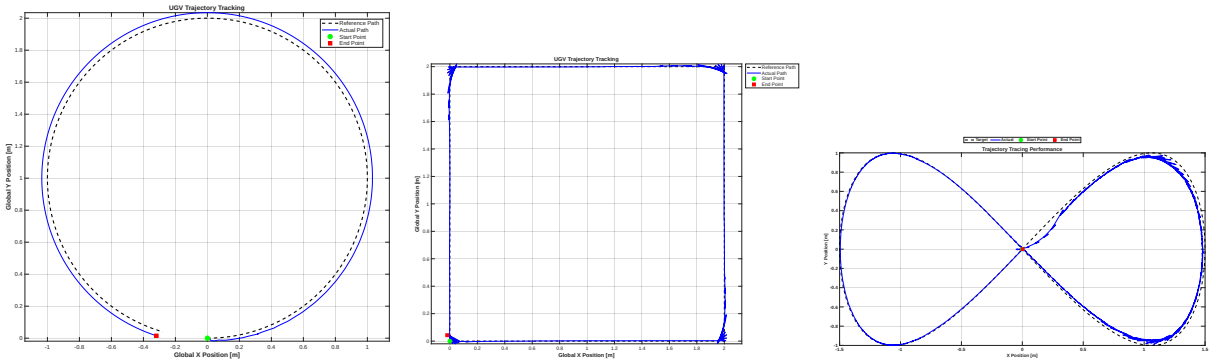
The adaptive backstepping controller improved tracking accuracy and transient smoothness compared to the Lyapunov-based approach. Adaptive instantaneous centre of rotation (ICR) compensation improved skid-steer maneuverability under lateral slip, reducing lateral tracking errors and wheel velocity oscillations. The LESO and STA-SMC further enhanced robustness against unmodeled dynamics and wheel–ground interaction uncertainties. In Fig 6a tracking performance degraded along highly discontinuous trajectories due to the absence of predictive optimization.

0.6.3 NMPC + ADRC + LESO

The proposed NMPC + ADRC + LESO architecture achieved the best overall tracking performance among the evaluated methods. NMPC generated smooth, dynamically feasible velocity commands while accounting for system constraints and future trajectory evolution, enabling anticipatory responses during sharp turns and discontinuous path transitions. The LESO estimated lumped disturbances in real time, while the inner-loop ADRC compensated for wheel slip, friction variations, contact nonlinearities, and unmodeled terrain effects. Reference acceleration feedforward further reduced tracking lag and improved wheel velocity convergence during aggressive maneuvers. The architecture was evaluated across different reference trajectories, as illustrated in Fig. 5.

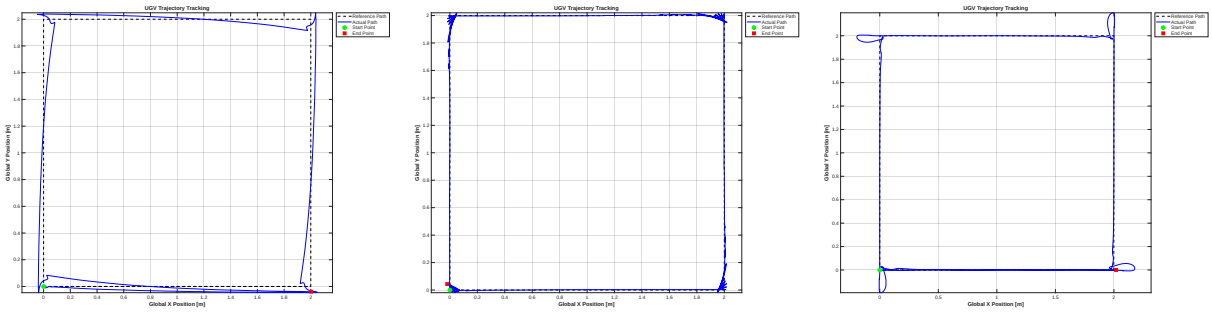
0.6.4 Robustness Under Discontinuous Trajectories

To evaluate controller robustness, all architectures were tested on discontinuous square trajectories involving abrupt direction changes and sharp heading transitions Fig 6.
The Lyapunov-based controller exhibited noticeable overshoot and oscillatory behavior during corner switching. The backstepping controller improved corner smoothness but still experienced transient delay under rapid heading reversals.



(a) Circular (b) Square (c) Figure-eight

Figure 5: Trajectory tracking performance of the proposed NMPC-LESO-ADRC controller under circular, square, and figure-eight reference trajectories.



(a) Backstepping (b) NMPC+ADRC (c) Lyapunov

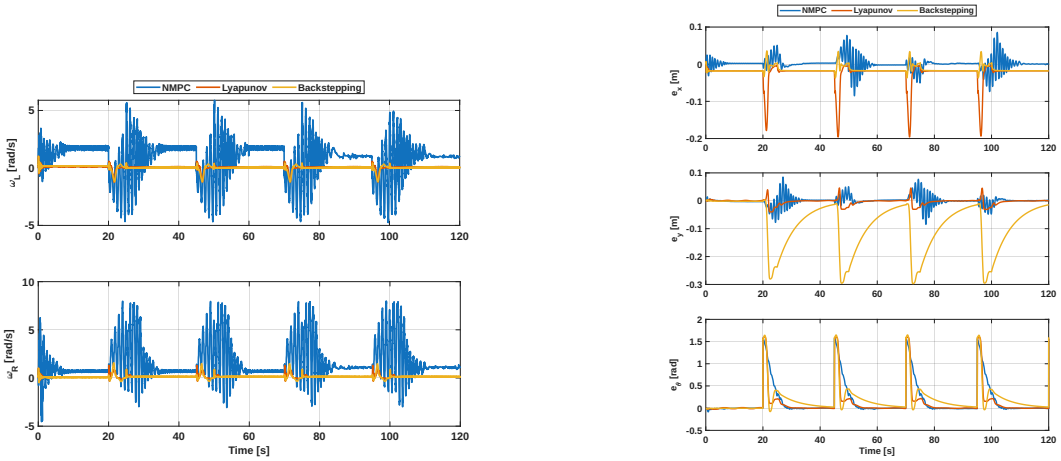
Figure 6: Trajectory tracking performance of the proposed controllers under square reference trajectories.

In contrast, the NMPC + ADRC architecture maintained stable trajectory tracking with significantly reduced overshoot and smoother velocity transitions. The predictive optimization capability enabled the controller to anticipate discontinuities before they occurred, resulting in superior path-following performance.

0.6.5 Tracking Performance

The trajectory tracking performance and error metrics are illustrated in Fig 7b and Table 3 respectively. The NMPC controller achieves the best overall tracking by simultaneously optimizing all tracking errors over a finite prediction horizon, resulting in balanced error correction. In contrast, the Lyapunov-SMC and Backstepping-SMC controllers rely on instantaneous error feedback, causing larger transient deviations during abrupt changes in trajectory direction.

Fig. 5 shows that the NMPC controller closely follows both the square and lemniscate trajectories, particularly through sharp corners and high-curvature regions. By optimizing the control inputs over a finite prediction horizon while accounting for system constraints, NMPC anticipates changes in the reference trajectory and adjusts the wheel velocities



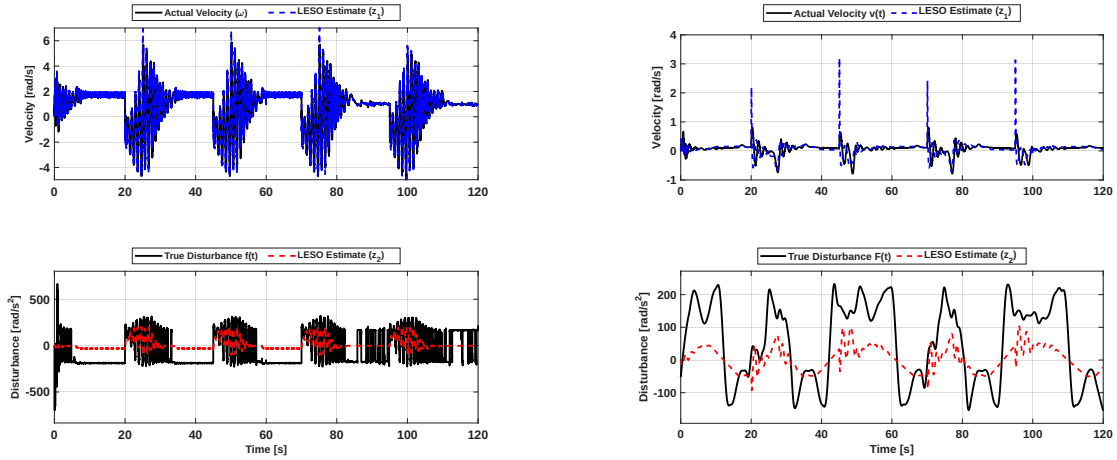
(a) Wheel Velocity Comparison

(b) Tracking Error Comparison

Figure 7: Comparison of wheel velocities and trajectory tracking errors for the evaluated controllers.

Controller	RMSE$_x$ (m)	RMSE$_y$ (m)	RMSE$_\theta$ (rad)	Max $\lvert e_x \rvert$ (m)	Max $\lvert e_y \rvert$ (m)	Max $\lvert e_\theta \rvert$ (rad)
NMPC	**0.0165**	0.0165	0.4134	0.0855	0.0852	**1.5720**
Lyapunov-SMC	0.0386	**0.0113**	**0.3681**	0.1959	**0.0462**	1.6074
Backstepping-SMC	0.0171	0.1231	0.3874	**0.0376**	0.2969	1.6489

Table 3: Trajectory Tracking Performance Comparison over a 120 s Simulation Horizon.



(a) LESO+ADRC+NMPC Disturbance and State Estimation

(b) Hybrid ADRC-SMC Disturbance and State Estimation

Figure 8: Comparison of disturbance and state estimation performance for the evaluated control architectures.

proactively. Consequently, the wheel velocity responses in Fig. 7a are smoother and more coordinated. In contrast, the Lyapunov-SMC and Backstepping-SMC controllers generate more abrupt wheel velocity variations.

0.6.6 Disturbance and State Estimation

Figures 8a and 8b present the state and disturbance estimation results of the observer-based dynamic control layers. In both architectures, the observers accurately estimate the wheel velocity states with fast convergence and negligible steady-state error. The observer successfully captures the dominant disturbance dynamics required for effective compensation. In the NMPC-ADRC architecture, the disturbance estimate is utilized for online disturbance rejection, whereas in the Lyapunov and Backstepping architectures the second-order LESO operates in conjunction with the hybrid ADRC-SMC, whose inherent robustness compensates for residual estimation errors. Consequently, both dynamic controllers maintain stable velocity tracking under wheel-ground interaction effects, model uncertainties, and external disturbances.

0.6.7 Discussion

The comparative analysis demonstrates that robust skid-steer trajectory tracking cannot rely solely on classical linear control methods under realistic terrain interaction conditions.The Lyapunov and backstepping approaches provide mathematically stable nonlinear tracking performance and exhibit strong robustness under moderate disturbances. However, their reactive nature limits performance under highly nonlinear and discontinuous trajectories.This integration enables accurate and stable tracking even in the presence of nonlinear slip, actuator saturation, and uncertain terrain interaction.Overall, the simulation results demonstrate that the proposed cascaded NMPC + ADRC + LESO framework provides a significant improvement in trajectory tracking accuracy, robustness, and smoothness compared to conventional nonlinear tracking strategies for skid-steer mobile robots. Future work will focus on experimental validation on a physical skid-steer platform, advanced disturbance estimation and compensation for complex terrain interactions, adaptive parameter tuning, and real-time implementation for autonomous outdoor navigation.

Appendix A

STABILITY ANALYSIS

This section establishes the closed-loop stability of the proposed cascaded NMPC-LESO-ADRC framework. The analysis proceeds by first establishing the input-to-state stability (ISS) of the LESO estimation error dynamics, followed by the ISS property of the constrained NMPC tracking dynamics. These results are subsequently combined through a composite Lyapunov function to establish the Uniform Ultimate Boundedness (UUB) of the overall closed-loop system.

A.0.1 Mathematical Preliminaries

Since the constrained NMPC value function is locally Lipschitz and may become non-differentiable due to constraint activation, the classical Lyapunov derivative is replaced by the *Upper Dini derivative*.

Let $V : \mathbb{R}^n \to \mathbb{R}$ be locally Lipschitz. Its Upper Dini derivative is defined as

$$D^+V(x) = \limsup_{h\to 0^+} \frac{V(x(t+h)) - V(x(t))}{h}. \tag{A.1}$$

For locally Lipschitz functions V_1 and V_2, the following subadditivity property holds:

$$D^+(V_1 + \theta V_2) \leq D^+V_1 + \theta D^+V_2, \quad \theta > 0, \tag{A.2}$$

which is used in the composite Lyapunov stability analysis.

The following assumptions are adopted.

Assumption 1: The NMPC optimization problem remains recursively feasible, and the optimal value function $V_{\mathrm{NMPC}}(e)$ is positive definite and locally Lipschitz [13].

Assumption 2: The lumped disturbance $f(x, u, t)$ is continuously differentiable over the operating region, satisfying

$$\|\dot{f}\| \leq M_0 + L_{f1}\|e\| + L_{f2}\|\tilde{z}\|, \tag{A.3}$$

where $\tilde{z}$ denotes the LESO estimation error.

Assumption 3: The observer bandwidth ω_o is selected such that the LESO estimation error dynamics are exponentially stable in the absence of disturbance derivative terms.

A.0.2 Stability of the LESO Estimation Error

The first-order wheel dynamics are (**??**)

The second-order LESO employed in the proposed ADRC framework **??**
where $\beta_1 = 2\omega_o$ and $\beta_2 = \omega_o^2$, where ω_o is the observer bandwidth.
Define the observer estimation errors

$$\tilde{v} = \omega - \hat{\omega}, \qquad \tilde{f} = f - \hat{f}. \tag{A.4}$$

Subtracting the observer dynamics from the plant dynamics yields

$$\dot{\tilde{v}} = \tilde{f} - \beta_1 \tilde{v}, \tag{A.5}$$

$$\dot{\tilde{f}} = \dot{f} - \beta_2 \tilde{v}. \tag{A.6}$$

Defining the estimation error vector $\tilde{z} = [\,\tilde{v} \;\; \tilde{f}\,]^T$, the error dynamics become
the error dynamics can be expressed compactly as

$$\dot{\tilde{z}} = A_o \tilde{z} + B_o \dot{f}, \tag{A.7}$$

where

$$A_o = \begin{bmatrix} -\beta_1 & 1 \\ -\beta_2 & 0 \end{bmatrix}, \qquad B_o = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Since$\beta_1 = 2\omega_o$,$\beta_2 = \omega_o^2$
the characteristic polynomial of A_o becomes

$$(s + \omega_o)^2,$$

which is Hurwitz for every $\omega_o > 0$.
Choose the quadratic Lyapunov function

$$V_o = \tilde{z}^T P \tilde{z}, \qquad P = P^T > 0, \tag{A.8}$$

Differentiating (A.8) along the observer error dynamics gives

$$\dot{V}_o = -\tilde{z}^T Q \tilde{z} + 2\tilde{z}^T P B_o \dot{f}. \tag{A.9}$$

Applying Young's inequality,

$$2a^T b \leq \varepsilon \|a\|^2 + \frac{1}{\varepsilon}\|b\|^2,$$

yields

$$\dot{V}_o \leq -\lambda_{\min}(Q)\|\tilde{z}\|^2 + \varepsilon\|\tilde{z}\|^2 + \frac{\|PB_o\|^2}{\varepsilon}\|\dot{f}\|^2. \tag{A.10}$$

Selecting

$$0 < \varepsilon < \lambda_{\min}(Q),$$

gives

$$\dot{V}_o \leq -c_1\|\tilde{z}\|^2 + c_2\|\dot{f}\|^2, \tag{A.11}$$

where

$$c_1 = \lambda_{\min}(Q) - \varepsilon, \qquad c_2 = \frac{\|PB_o\|^2}{\varepsilon}.$$

Therefore, the LESO estimation error dynamics are Input-to-State Stable (ISS) with respect to the disturbance derivative $\dot{f}$. Furthermore, increasing the observer bandwidth ω_o improves the estimation convergence rate while maintaining bounded estimation errors.

A.0.3 Input-to-State Stability of the NMPC Closed-Loop System

Consider the body-frame tracking error dynamics

$$\dot{e} = f(e, u) + w, \tag{A.12}$$

where $e = [x_e, y_e, \theta_e]^T$ is the tracking error, $u = [v, \omega]^T$ is the NMPC control input, and $w = L_v\tilde{z}$ denotes the disturbance induced by the LESO estimation error.. Here, $L_v > 0$ represents the maximum finite Lipschitz constant of the piecewise smooth NMPC control law evaluated over the compact feasible operating domain.

The NMPC computes the optimal control sequence by minimizing the finite-horizon cost

$$J^*(e) = \min_{u(\cdot)} \left(\int_0^T \ell(e, u)\, dt + V_f(e(T)) \right), \tag{A.13}$$

subject to the system constraints. Under standard NMPC assumptions, the optimal value function $V_{\text{NMPC}}(e) = J^*(e)$ is positive definite and locally Lipschitz. Hence, for the nominal system,

$$D^+ V_{\text{NMPC}} \leq -c_3\|e\|^2, \qquad c_3 > 0. \tag{A.14}$$

With the disturbance $w = L_v\tilde{z}$,

$$D^+V_{\text{NMPC}} \leq -c_3\|e\|^2 + L_v\|e\|\|\tilde{z}\|. \tag{A.15}$$

Applying Young's inequality,

$$L_v\|e\|\|\tilde{z}\| \leq \frac{\varepsilon}{2}\|e\|^2 + \frac{L_v^2}{2\varepsilon}\|\tilde{z}\|^2,$$

yields

$$D^+V_{\text{NMPC}} \leq -\left(c_3 - \frac{\varepsilon}{2}\right)\|e\|^2 + \frac{L_v^2}{2\varepsilon}\|\tilde{z}\|^2. \tag{A.16}$$

Choosing $0 < \varepsilon < 2c_3$ and defining

$$c_4 = c_3 - \frac{\varepsilon}{2}, \qquad c_5 = \frac{L_v^2}{2\varepsilon},$$

gives

$$D^+V_{\text{NMPC}} \leq -c_4\|e\|^2 + c_5\|\tilde{z}\|^2, \tag{A.17}$$

which establishes the ISS property of the NMPC subsystem with respect to the LESO estimation error.

A.0.4 Composite Stability of the Cascaded NMPC-LESO-ADRC Framework

The individual ISS properties established for the LESO estimation error dynamics and the NMPC tracking subsystem are now combined to analyze the stability of the complete cascaded control architecture.

Consider the composite Lyapunov function

$$V(e, \tilde{z}) = V_{\text{NMPC}}(e) + \theta V_o(\tilde{z}), \qquad \theta > 0, \tag{A.18}$$

Since both functions are positive definite and radially unbounded, the composite function $V(e, \tilde{z})$ is also positive definite and radially unbounded.

Using the subadditivity property of the Upper Dini derivative,

$$D^+V \leq D^+V_{\text{NMPC}} + \theta\dot{V}_o,$$

and substituting (A.11) and (A.17) yields

$$\begin{aligned} D^+V \leq & -c_4\|e\|^2 + c_5\|\tilde{z}\|^2 \\ & - \theta c_1\|\tilde{z}\|^2 + \theta c_2\|\dot{f}\|^2. \end{aligned} \tag{A.19}$$

Choosing $\theta > \frac{c_5}{c_1}$ and defining $c_6 = \theta c_1 - c_5 > 0$ gives

$$D^+V \leq -c_4\|e\|^2 - c_6\|\tilde{z}\|^2 + \theta c_2\|\dot{f}\|^2. \tag{A.20}$$

and consequently,

$$D^+V \leq -c_4\|e\|^2 - c_6\|\tilde{z}\|^2 + \theta c_2\|\dot{f}\|^2. \tag{A.21}$$

From Assumption 2, the disturbance derivative is bounded, i.e.,

$$\|\dot{f}\| \leq \delta,$$

for some positive constant δ. Therefore,

$$D^+V \leq -c_4\|e\|^2 - c_6\|\tilde{z}\|^2 + \theta c_2\delta^2. \tag{A.22}$$

Equation (A.22) shows that the composite Lyapunov function decreases outside a compact neighborhood of the origin.

Under Assumptions 1–3, the proposed cascaded NMPC–LESO–ADRC controller guarantees that all closed-loop signals remain bounded. Moreover, the tracking error e and the LESO estimation error $\tilde{z}$ are uniformly ultimately bounded (UUB).

From (A.22), the negative definite terms dominate the bounded disturbance term whenever

$$c_4\|e\|^2 + c_6\|\tilde{z}\|^2 > \theta c_2\delta^2, \tag{A.23}$$

which implies

$$D^+V < 0.$$

Hence, every closed-loop trajectory enters, in finite time, the compact invariant set

$$\Omega = \left\{(e, \tilde{z}) : c_4\|e\|^2 + c_6\|\tilde{z}\|^2 \leq \theta c_2\delta^2\right\}. \tag{A.24}$$

Once inside Ω, the trajectories remain bounded for all future time. Therefore, both the tracking error e and the LESO estimation error $\tilde{z}$ are uniformly ultimately bounded, completing the proof.

Bibliography

[1] D. Pazderski and K. Kozłowski, "Trajectory tracking control of skid-steering robot–experimental validation," *IFAC Proceedings Volumes*, vol. 41, no. 2, pp. 3106–3111, 2008.

[2] L. S. Vilas Boas, T. L. M. Santos, and A. G. S. Conceição, "A simplified trajectory tracking control based on linear design for skid-steered wheeled ugvs," *Journal of the Franklin Institute*, vol. 361, no. 4, p. 106660, 2024. [Online]. Available: https://doi.org/10.1016/j.jfranklin.2024.106660

[3] S. Rabiee and J. Biswas, "A friction-based kinematic model for skid-steer wheeled mobile robots," in *2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 8563–8569.

[4] Y. Chen, N. Li, W. Zeng, S. Zhang, and G. Ma, "Curved path following controller for 4w skid-steering mobile robots using backstepping," *IEEE Access*, vol. 10, pp. 66 072–66 082, 2022.

[5] Z. Jian, W. Shuang-shuang, L. Hua, and L. Bin, "The sliding mode control based on extended state observer for skid steering of 4-wheel-drive electric vehicle," in *2012 2nd International Conference on Consumer Electronics, Communications and Networks (CECNet)*, 2012, pp. 2195–2200.

[6] T. Nascimento, C. Dorea, and L. Gonçalves, "Nonholonomic mobile robots' trajectory tracking model predictive control: a survey," *Robotica*, vol. 36, pp. 1–21, 01 2018.

[7] J. Rawlings and D. Mayne, *Model Predictive Control: Theory and Design*, 01 2009.

[8] A. Javier Prado, D. Chávez, O. Camacho, M. Torres-Torriti, and F. Auat Cheein, "Adaptive nonlinear mpc for efficient trajectory tracking applied to autonomous mining skid-steer mobile robots," in *2020 IEEE ANDESCON*, 2020, pp. 1–6.

[9] C. Zhu, E. Hill, M. Biglarbegian, S. Gadsden, and J. Cline, "Smart agriculture: Development of a skid-steer autonomous robot with advanced model predictive controllers," *Robotics and Autonomous Systems*, vol. 162, p. 104364, 04 2023.

[10] L. Li, W. Cao, H. Yang, and Q. Geng, "Trajectory tracking control for a wheel mobile robot on rough and uneven ground," *Mechatronics*, vol. 83, p. 102741, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957415822000034

[11] J. Han, "From pid to active disturbance rejection control," *IEEE Transactions on Industrial Electronics*, vol. 56, no. 3, pp. 900–906, 2009.

[12] J.-Y. Jun, M.-D. Hua, and F. Benamar, "A trajectory tracking control design for a skid-steering mobile robot by adapting its desired instantaneous center of rotation," in *53rd IEEE Conference on Decision and Control*, 2014, pp. 4554–4559.

[13] D. Teichrib and M. S. Darup, "Error bounds for maxout neural network approximations of model predictive control," *IFAC-PapersOnLine*, vol. 56, no. 2, p. 10113–10119, 2023. [Online]. Available: http://dx.doi.org/10.1016/j.ifacol.2023.10.883